ARGA BONA IMMANUEL NAINGGOLAN

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Recent Computer Technology graduate from the Del Institute of Technology with a strong foundation in integrated computer systems. Highly passionate about Web Development, Internet of Things (IoT), Cloud infrastructure, and Data. Adept at bridging hardware and software to create efficient, visually appealing, and intelligent technological solutions. A dedicated lifelong learner eager to leverage technical expertise across these domains to solve real-world problems.

Projects

Web Development Project - Genshin Impact Character Database Website using REST API May 2024 - Jun 2024

- Implement REST API to display character data dynamically and in real time on the website.

Embedded System Project - Lampu Otomatis Menyala Ketika Ada Pergerakan Oct 2024 - Dec 2024

- Design and implement an automatic lighting system based on a motion sensor (PIR Sensor) that can turn on the lights automatically when it detects movement around the surveillance area.

Internet of Things Project - Smart Door Lock And Access Oct 2025 - Dec 2025

- Designing a safe, efficient, and integrated automatic lock system with IoT and implementing RFID/fingerprint-based user authentication features with fast and accurate responses.

Final Project 1 - Layanan Kesehatan SIBALTAR (Siantar, Balige, Dan Tarutung) Jan 2024 - Jun 2024

- Designed and implemented a web-based health service information system for three regions (Siantar, Balige, Tarutung). and display details of hospitals, polyclinics, and doctors in a structured and easy-to-understand manner and increase the efficiency of public health information access through a centralized digital platform.

Final Project 2 - Sistem Pemberi Pakan Ayam Berbasis Internet of Things (IoT) Jan 2025 - Jun 2025

- designing an automatic feeding and watering system that can be controlled via an application and monitored in real time.

Final Project - Sistem Pemantauan Tinggi Permukaan Air Danau Toba Berbasis IoT Menggunakan Teknologi LoRa Aug 2025 - Aug 2026

- Designing and implementing a real-time water level monitoring system for Lake Toba based on the Internet of Things (IoT) using LoRa communication.

Intern Project - Penerapan Python-Based Automation dalam Pemetaan Desain Spasial KML Menjadi Bill of Quantities (BOQ) Jaringan FTTH Telkom - **PT Telkom Infrastruktur Indonesia** Feb 2026 - Jun 2026
Medan, Sumatera Utara

- Developed the FTTH BOQ Generator web application (Python, Streamlit) for PT Telkom Indonesia, automating Bill of Quantities generation for Fiber-to-the-Home networks from Google Earth Pro KML files

Jenjang Pendidikan

Institut Teknologi Del - **Jl. Sisingamangaraja Sitoluama-Laguboti. Toba Samosir 22381. Sumatera Utara-Indonesia** Aug 2023 - Sep 2026

Diploma in Computer Technology, 3.03/4.00

Pengalaman organisasi

Computer Technology Student Association - **Del Institute of Technology** Sep 2024 - Jun 2025

Head of The Publication Division

Managing all publication, documentation, and digital branding activities for the student association. Coordinate the creative team in producing informative and engaging content to enhance the organization's image, and ensure all activity information is effectively communicated to students and the public.

Keterampilan, Prestasi & pengalaman lain

- **Modules Taken** : Computer Architecture and Organization, Digital Systems and Microprocessors, Embedded Systems and IoT Applications, Computer Networks and Network Security, Operating Systems, Database Systems and Data Management, Software Engineering, Object-Oriented Programming, Web Programming and Development, Algorithms and Data Structures, Computer Systems and Maintenance, Industrial Automation and Control Systems, Cloud Computing and Virtualization, Project Management and Professional Ethics
- **Soft Skills**: Teamwork & Collaboration, Communication & Presentation Skills, Critical Thinking & Problem Solving, Time Management, Leadership
- **Programming Languages**: C, C++, HTML, CSS, PHP, JavaScript, SQL, Python
- **Software**: VS Code, Arduino IDE, Cisco Packet Tracer, VirtualBox, Linux / Ubuntu
- **Hardware**: Arduino, Microcontroller, LoRa Communication, IoT, Embedded System